Experiment No : 2

Title: Perform recursive binary search using C.

Name of Student: Shamiksha Chaudhari
Class: SY **Div:** SY1 **PRN No:** B25ET1005 **Batch:** A1

Program:

```
#include <stdio.h>
int binarysearch(int a[], int l, int r, int target);
int main()
{
   int a[10], n, target, result ,i, m ;
   printf("enter elements of the array : ");
   for (i=0;i<10;i++)
   {
      scanf("%d" , &a[i]);
   }
   n = sizeof((a)) / sizeof a[0];
   printf("enter element to be searched : ");
   scanf("%d" , &target);
   result=binarysearch(a , 0, n-1, target);
   if (result != 1)
     printf("Targeted element %d found in the array at position %d " ,target, result);
   else
     printf("Targeted element not found in the array");
}

int binarysearch(int a[], int l , int r, int target)
{
   if (l > r)
   {
     return 0;
   }
   int mid = l + ((r - l)/2);

   if (a[mid] == target)
   {
     return mid;
```

```
    }
    else if (a[mid] >target)
    {
      return binarysearch(a ,l, mid-1 , target);
    }
    else
    {
        return binarysearch(a, mid + 1, r, target);
    }

}
```

Output :

Output

```
enter elements of the array : 1 2 3 4 5 6 7 8 9 10
enter element to be searched : 7
Targeted element 7 found in the array at position 6

=== Code Execution Successful ===
```